The Weak Temperature Gradient: A review of the possible implementations in small-domain models

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Key Points:

- Radiative-Convective equilibrium under the Weak Temperature Gradient framework is unstable to perturbations
- This instability is consistent over a range of parameter spaces
- The nature of this instability is dependent on vertical resolution

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Abstract

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1 Introduction

The weak temperature gradient (WTG) approximation (Sobel & Bretherton, 2000) is a simplified framework for atmospheric dynamics in the deep tropics where the Coriolis force is weak. In such a framework, buoyancy gradients in the free troposphere are rapidly smoothed out by gravity waves, and thus spatial temperature gradients in the free troposphere and their evolution over time are considered small enough to be ignored. Local perturbations in buoyancy caused by heating (cooling) are therefore assumed to be balanced by vertical ascent (subsidence). Thus, vertical motion is strongly coupled to convection within the deep tropics, rather than it being a casual relationship (Raymond & Zeng, 2005). Therefore, when we run single-column models (SCMs) or small-domain cloud-resolving model (CRM) simulations to simulate tropical processes, the WTG approximation provides a more suitable framework for the parameterization of the large-scale circulation and its coupling with convective activity then directly specifying the large-scale vertical ascent as a forcing.

There are a number of studies (e.g., Raymond & Zeng, 2005; Sobel et al., 2007; Sessions et al., 2010; Daleu et al., 2012; Emanuel et al., 2014; Daleu et al., 2015, and others) that investigate the application the WTG approximation framework to SCM and CRM simulations that were initially spun-up to Radiative-Convective Equilibrium (RCE). The common feature about most of these studies is that applying the WTG approximation to these simulations causes a bifurcation in the model state into either (1) a dry, often non-precipitating state, or (2) a much moister state with heavy precipitation. Emanuel et al. (2014) in particular deduced that two observed possible model states are actually analogues to the wet and dry states seen in large-domain RCE simulations when convection undergoes large-scale self-aggregation (Fig. 1a). Under the WTG approximation, they found that above a certain "critical" sea-surface temperature the RCE state is actually in unstable equilibrium, where even the slightest deviation from the RCE state

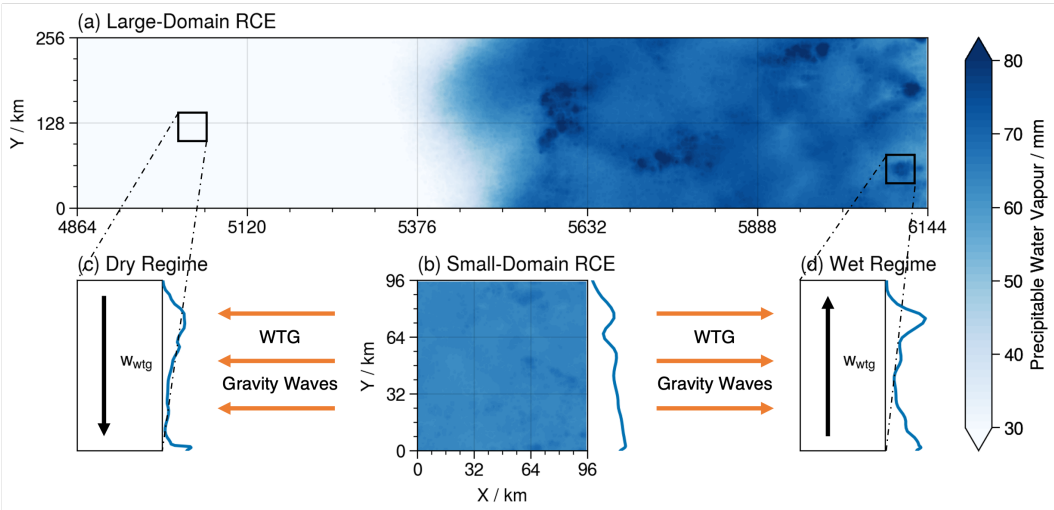

Figure 1. When (a) a large-domain is run to radiative-convective equilibrium (a), we see that the model separates into wet and dry regimes. This can be using (b) small-domain simulations, which are shown to be unstable under the WTG framework, collapsing into either (c) a dry regime with subsidence (note that $w_{\mathrm{wtg}} < 0$), or (d) a wet regime where $w_{\mathrm{wtg}} > 0$, thus enhancing adiabatic ascent and low-level moisture. The blue curves are the average relative-humidity profiles of (b) the entire domain, and the 1000 columns with the (c) lowest and (d) highest column relative humidities respectively.

(Fig. 1b) when the WTG approximation is applied will cause the model to collapse into one of the two regimes (Fig. 1c,d).

Furthermore, these studies also showed that the degree to which this bifurcation occurs depends on the model setup, the WTG scheme that was used, and especially on sea-surface temperature and the radiation scheme used. In particular, many studies where the radiation scheme was not interactive, but instead of a form similar to that of Pauluis and Garner (2006) where instead a fixed cooling rate of 1.5 K day^{-1} was imposed across most of the free troposphere and a relaxation of temperature towards 200 K in the rest of the atmosphere, showed a muted bifurcation (or moistening or drying) in the possible model states (Raymond & Zeng, 2005; Daleu et al., 2015) that resulted when the WTG approximation was applied.

These results emphasize the necessity of quantifying the degree of instability caused by implementing the WTG approximation in a model spun up to RCE. However, while the abovementioned studies do investigate the impact of applying the WTG approximation to SCM or CRM simulations, many of the CRM simulations were conducted in 2D, instead of 3D, necessitating that the horizontal winds be relaxed to zero in order to prevent a spontaneous circulation to develop within the domain and the self-aggregation of convection. More importantly, these studies emphasized and explored the bifurcation of the model state as a dependency on the initial model conditions, rather than the parameters of the WTG approximation implementation. For example, the method that implements the WTG approximation which is most mainstream are derivatives of the method used in Raymond and Zeng (2005), which at its root relates the vertical ascent due to the weak-temperature approximation w_{WTG} to the difference between the horizontal average of the domain potential temperature $\overline{\theta}$ to the large-scale potential temperature θ_{LS} as follows:

$$w_{\mathrm{WTG}} \propto \frac{\overline{\theta} - \theta_{\mathrm{LS}}}{\tau} \tag{1}$$

where τ is a relaxation constant.

However, in many of the abovementioned studies, τ is rarely varied, even though it is an important component in determining the strength of the WTG approximation forcing on the model. All the different possible WTG schemes that could be implemented have analogues to τ that determine the strength of the WTG approximation and could therefore vastly impact experimental results if they are not factored in. Therefore, in this study, we systematically investigate the impact that varying the strength at which the WTG approximation is applied to the model can have on models in RCE over a swathe of sea-surface temperatures and radiative schemes.

In Section 2, I elaborate on the 3D-CRM I used and the different WTG schemes that I used, as well as the different experimental configurations that I ran to investigate the impact that these WTG schemes can have. In Section 3, I cover the procedures I used in the RCE spinup for our experiments, and present the different RCE states that result from the different experimental configurations in Section 2. In Section 4, I then proceed to apply the WTG approximation at different forcing strengths to the models that have reached the RCE state, and present my analysis of the bifurcation states that result, as well as their implications for future experiments. I summarize and conclude in Section 5.

2 Weak Temperature Gradient implementations in small-domain Cloud-Resolving Models

Since the WTG approximation was conceptualized by Sobel and Bretherton (2000), there have been at least two different implementations of the WTG that have been created to parameterize the large-scale circulation, or specifically, to couple the large-scale vertical motion to the perturbations in large-scale buoyancy profile:

(1) The Weak Temperature Gradient (WTG) implementation, which directly calculates large-scale vertical motion from the perturbations in the large-scale potential temperature profile.
(2) The Damped Gravity Wave (DGW) implementation, which uses damped and linearized momentum equations to relate the large-scale vertical motion from the perturbations in the large-scale virtual temperature profile.

Each of these implementations calculates the vertical motion due to the WTG approximation w_{wtg}. This vertical motion is responsible for advecting and mixing the temperature and water vapour profiles of the CRM domain with the large-scale profiles. As such, because ascent is accompanied by advection of moisture into the domain, and subsidence causes advection of moisture out of the domain, this calculated w_{wtg} is critical in moistening or drying out the model simulation. *Note: the increase in relative humidity is partially due to overall advection of water vapour into the domain, and also because of the fact that adiabatic ascent of an atmospheric parcel caused by w_{wtg} causes, by the Clausius-Clapeyron equation, the relative humidity to increase because the temperature drops.*

In this section, we go through the different individual implementations, the parameters that affect the strength of their implementation, as well as discuss each of their known drawbacks.

2.1 The Weak Temperature Gradient Implementation

This implementation assumes that the vertical advection of virtual potential temperature $w\partial_z\theta_v$ removes the differences in the buoyancy between the CRM and large-scale environment over a time-scale τ, such that at a height in the free troposphere z_i:

$$w_{\mathrm{wtg}}(z_i)\frac{\partial\overline{\theta}}{\partial z}\bigg|_{z=z_i} = \frac{\overline{\theta}(z_i)-\theta_0(z_i)}{\tau}\sin\frac{\pi z_i}{h} \tag{2}$$

This method of implementation in a CRM was first done by (Raymond & Zeng, 2005), and has since become the most popular method of implementing the WTG approximation due to the straightforward conceptual picture it provides. It is noted of course, that the presence of the relaxation timescale τ indicates that there will always be a time-mean perturbation in the temperature compared to the large-scale, such that the WTG is not perfectly obeyed. Nonetheless, we would expect that as τ decreases, the degree to which the WTG approximation must be obeyed in the troposphere must increase, which is what we indeed find in our experimental results in Section 4.

2.2 The Damped Gravity Wave Implementation

At their core, the DGW implementations are based on the damping of gravity waves by the Rayleigh damping coefficient a_m in the momentum equations (3) and (4):

$$u'_t = -\frac{1}{\rho}p'_x + fv - a_m u' \tag{3}$$

$$v'_t = -\frac{1}{\rho}p'_y - fu - a_m v' \tag{4}$$

Using the ideal gas law, hydrostatic balance and mass conservation laws, the momentum equations are thus transformed into the following governing equation:

$$\frac{\partial}{\partial p}\left(\frac{f^2 + a_m^2}{a_m}\frac{\partial \omega'}{\partial p}\right) \approx \frac{k^2 R_d}{p} T_v' \tag{5}$$

where R_d is the dry gas constant, T_v is the virtual temperature, and k is the horizontal wavenumber of the gravity wave. $(\cdot)'$ represents the perturbation of $(\cdot)$ from the large-scale mean-state. Eq. 5 is the implementation that is currently used in the System for Atmospheric Modelling, as done by Blossey et al. (2009). Kuang (2008) also implemented a similar form in the SAM using vertical coordinates instead of pressure coordinates. Eq. 5 reduces to that of Kuang (2008) upon setting the Coriolis parameter $f = 0$. Upon this assumption, which is valid in the deep tropics in which our model framework is conducted, and assuming that a_m is constant with height, we simplify Eq. 5 to:

$$\frac{\partial^2 \omega'}{\partial p^2} = \frac{k^2}{a_m}\frac{R_d T_v'}{p} \tag{6}$$

A comparison of Eq. (2) and (6) shows that a_m in the DGW framework is the analogue to τ in the WTG implementation. However, we can also modify the strength of the WTG implementation by changing the wavenumber k of the gravity wave under consideration. This is a drawback of using the DGW implementation over the WTG implementation: it can only consider one horizontal wave-number at any time. However, the DGW implementation produces w_{wtg} curves in the troposphere that are smoother and less top-heavy than when the WTG implementation is used. Furthermore, Raymond and Zeng (2005) and Daleu et al. (2012) both noted that the DGW implementation performs better than the WTG method when the static stability is near zero. Lastly, the WTG method cannot be applied at the boundary layer, and requires the user to impose an artificial boundary layer top in order to enforce this crierion. As such, Daleu et al. (2015) concludes that the DGW methodology overall better captures the dynamics of the weak temperature gradient.

3 Model Spinup to Radiative-Convective Equilibrium

3.1 Model Description and Experimental Configurations

We used the System for Atmospheric Modelling (SAM) (Khairoutdinov & Randall, 2003) version 6.10.9 configured in Cloud-Resolving mode. The model solves the anelastic continuity, momentum, and tracer conservation equations, with total nonprecipitating water (vapor, cloud water, cloud ice) and total precipitating water (rain, snow, graupel) included as prognostic thermodynamic variables. Simulations are run in three dimensions, with the horizontal domain of our model ranging from 1-4 km spacing between each grid point so as to allow for cloud-permitting simulations. The boundaries are doubly periodic. There are 64 vertical levels in our model, with the vertical spacing increasing from 50 m at the boundary layer to around 500 m at the tropical tropopause, to a total height of ∼27 km with a rigid upper-bound. Damping is applied to the model top to reduce reflection of gravity waves. A simple Smagorinsky-type scheme is used for the effect of subgrid-scale motion. In all our experiments, the sea-surface temperature (SST) is spatially uniform and time-invariant, and there is no Coriolis force.

In our study, we varied the following model parameters:

- Sea-surface temperature: 295 K, 300 K and 305 K
- Domain size: (a) 64 × 64 km at 1 km res, (b) 128 × 128 km at 1,2 or 4 km res
- Vertical Levels: (a) 64 levels or (b) 28 levels
- Radiative Schemes: (a) RRTM Perpetual, (b) RRTM Diurnal, or (c) Pauluis and Garner (2006)
- Surface Fluxes: (a) Calculated, or (b) Bulk-Scheme using fixed surface wind (applicable only for Pauluis and Garner (2006) radiation)

The default reference model parameters are: (a) 128 × 128 km domain at (b) 2 km horizontal resolution with (c) 64 vertical levels, (d) a sea surface temperature of 300 K, (e) interactive RRTM scheme with perpetual insolation and (f) interactive surface fluxes .

3.2 Spinup to Radiative-Convective Equilibrium

Our first step is to spin up our model to radiative-convective equilibrium. However, the time taken for the model to reach a stable RCE depends on several factors: (1) the domain size, (2) horizontal and vertical resolutions, (3) sea-surface temperature and (4) the radiative scheme used. In order to account for internal model variability, we use a 10-member ensemble run, generated by creating perturbations to the initial model state, to generate equilibrium profiles.

Overall, we find that runs with interactive radiation (i.e. RRTM) require longer ($\sim \mathcal{O}(1000)$ days) to reach a stable RCE state compared to using a simple radiative scheme such as that of Pauluis and Garner (2006) ($\sim \mathcal{O}(200)$ days). We note that in many other experiments, the spinup time is 300 days or less, yet I have found that for small-domain cloud-permitting simulations the upper atmospheric temperature to reach a state of equilibrium at least 1000 days (and often much longer) of model spinup are needed. For example, with the default reference model parameters, we see that the model temporarily enters a state over the first 200 days whereby the upper troposphere and lower stratosphere are much colder and warmer respectively than their final equilibrium RCE states (Fig. 2). Following this, the model exits this state and slowly moves continuously back to an equilibrium RCE state. However, we see that though the lower tropospheric temperature reaches equilibrium pretty quickly, the upper atmospheric temperature requires $\sim \mathcal{O}(1000$ days) to reach equilibrium.

We consider a given temperature and humidity profile to have reached RCE state, when an ensemble model run spun-up from this combination of temperature and spe-

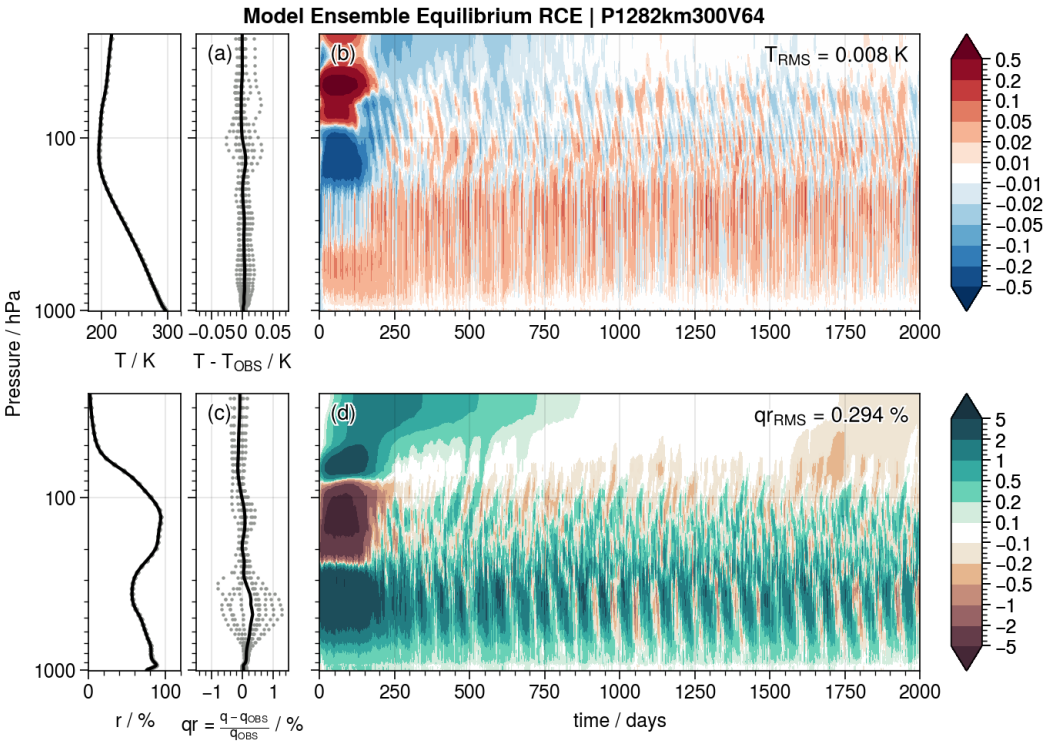

Figure 2.

cific humidity profiles returns the same vertical profile. Again using the default reference model parameters, we see that the final vertical profiles of temperature and specific humidity, averaged over the model ensemble, returns the initial profiles of temperature (T_{obs}) and specific humidity (q_{obs}), and therefore we consider this vertical profile to be in RCE. If the final temperature and specific humidity profiles are too far from T_{obs} and q_{obs}, there will be ramifications in our results in Section 4, as we will explain further in that section.

4 Applying the Weak Temperature Gradient framework to models in Radiative-Convective Equilibrium

Once the models have been spun-up to RCE, we take the vertical profiles of temperature and humidity as the large-scale reference profiles. We then apply the weak temperature gradient framework to the models over a range of τ or a_m (depending on the implementation scheme used) values, and run a 5-member ensemble over a period of 500 days for each of the configurations, taking statistics over the last 250 days. For each member in the ensembles, perturbations were made to the initial state of the model, resulting in a mix of wet and dry final states. Although other studies (e.g. Sobel et al., 2007; Sessions et al., 2010) force out the dry state by initializing the model runs with zero moisture, we have found that this is unnecessary. As long as the large-scale vertical profiles of temperature and humidity are very close to RCE, both the wet and dry regimes will appear.

For the DGW implementation, we vary the strength of the weak temperature gradient forcing on the model by varying the damping parameter a_m from 1 day^{-1} to 512 day^{-1}, with steps in exponents of 2. The wavelength of the gravity wave is taken to be 2600 km, which is the default implementation in Blossey et al. (2009). For simplicity, we assume that a_m is invariant in height/pressure.

4.1 A Gradual Implementation of the WTG Approximation

Implementing the Weak-Temperature Gradient approximation suddenly can cause a shock to the model, which will then transfer into a different regime state. Therefore, we implement a smooth transition from a pseudo-RCE state ($a_m \gg 1$) to a WTG state ($a_m = a_{m.0}$). This was by varying a_m in the form of an error function. This will allow the increase in the strength of the WTG approximation to taper off gently, instead of increasing more rapidly as it approaches $a_{m.0}$ (Fig. 3).

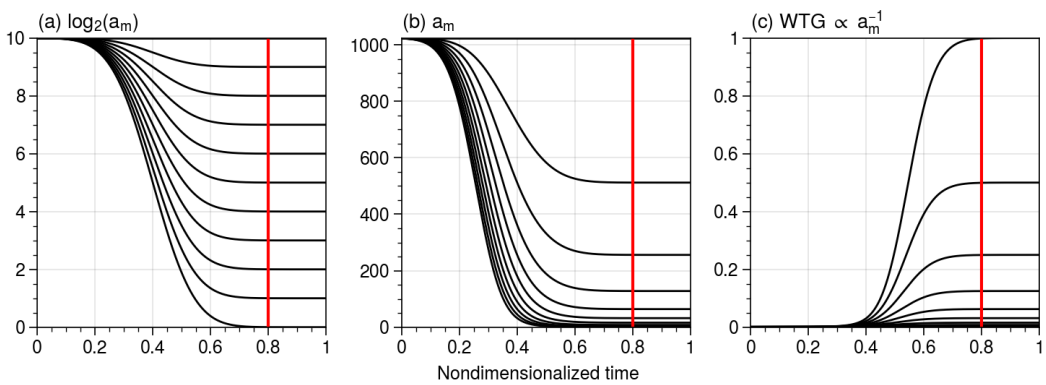

Figure 3. A visualization of how we scale a_m with time against a normalized scaling time t_{wtg} (red vertial line), where t_{wtg} represents the fraction of the total time for which large-scale forcing is applied, for the momentum damping a_m to reach its final value $a_{m.0}$.

The basic error function is given by

$$\operatorname{erf} t = \frac{2}{\sqrt{t}} \int_0^t e^{-x^2}\, \mathrm{d}x$$

We use the error function to vary the momentum damping parameter a_m, or more specifically, the parameter x where $x = \log_2 a_m$.

The formula for x as a function of time t, is given by

$$x(t) = \left(\frac{10 - \log_2 a_{m.0}}{2}\right) \operatorname{erf}\left(\frac{t_{\mathrm{wtg}}/2 - t}{t_{\mathrm{wtg}}/5}\right) + 5 + \frac{1}{2}\log_2 a_{m.0} \tag{7}$$

$$\therefore a_m(t) = \begin{cases} 2^{x(t)}, & t < t_{\mathrm{wtg}} \\ a_{m.0}, & t \geq t_{\mathrm{wtg}} \end{cases} \tag{8}$$

Where t_{wtg} is a scaling parameter, that defines the time when $a_m \approx a_{m.0}$ as a fraction of the total time that the WTG approximation is applied. So if the WTG approximation is applied over 100 days, then if $t_{\mathrm{wtg}} = 0.25$, $a_m \approx a_{m.0}$ occurs at day 25. When $t = 0$, $a_m(0) = 1024$, which we will see later is large enough such that w_{wtg} is negligible and does not produce any noticeable impact on the domain climatology. In our model, we set $t_{\mathrm{wtg}} = 0.2$, such that the WTG forcing in the model scales to its final strength when the model reaches 100 days. Although technically it is not necessary, we still allow for another 150 days for the model to adjust to the final WTG strength before taking statistics for the final 250 days of the model run.

4.2 A bifurcation in the model states is obtained

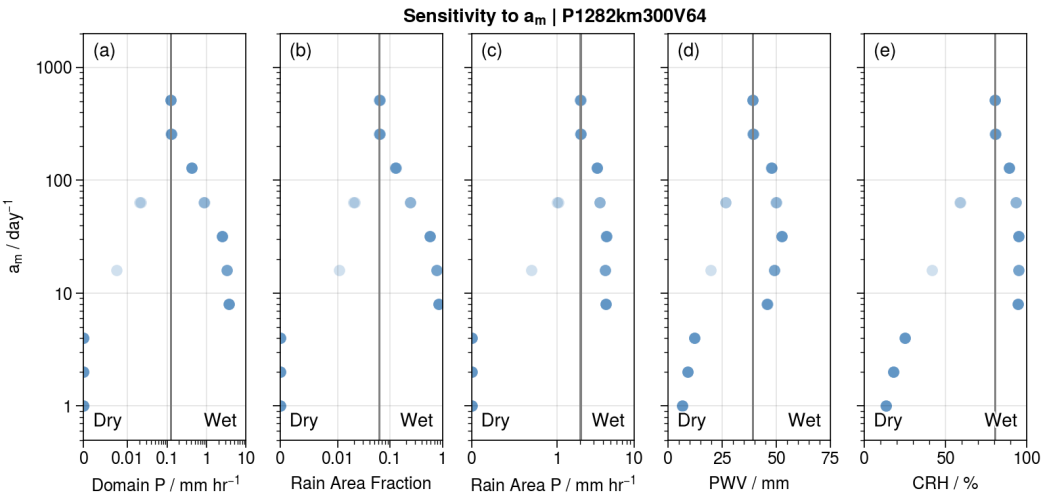

Figure 4. We plot (a) domain-mean precipitation, (b) rain area fraction, (c) rain area precipitation rate, (d) domain-mean precipitatable water vapour and (c) domain-mean column relative humidity that result when the Damped Gravity-Wave (DGW) framework (Kuang, 2008; Blossey et al., 2009) is applied with different strengths to a 10-member small-domain model ensemble under the P1282km300V64 configuration. As the strength of the DGW framework increases (i.e. $a_m \rightarrow 0$), a bifurcation in the final model state results, similar to what Emanuel et al. (2014) found in a single-column model.

Similar to Emanuel et al. (2014)'s results using the MITgcm in single-column mode, we found that applying the WTG framework to our small-domain models causes a bifurcation in the resultant model state (Fig. 4) into (1) dry (and even non-precipitating)

regimes, and (2) moist, strongly precipitating regimes. As the strength of the WTG approximation increases, the bifurcation becomes more pronounced. However, as the strength of the WTG approximation increases beyond a certain threshold (in Fig. 4 shown to be around $a_m \leq 4$ for the default configuration), the resultant model states collapse into a single non-precipitating regime.

This bifurcation is present across different sea surface temperatures, different horizontal resolutions and domain sizes, and even when the perpetually constant insolation is changed to a diurnal insolation. We plot the bifurcation for a configuration where the SST is 295 K as an example (Fig. 5). In this example, while it is apparent that there are minor differences in details (such as the fact that there is no lower limit to a_m that forces a dry state), the shape of the bifurcation curve is close to that of the control configuration.. To a lesser extent, this bifurcation is also present when the radiation is changed from an interactive scheme (RRTM) to a static scheme (e.g. Pauluis & Garner, 2006).

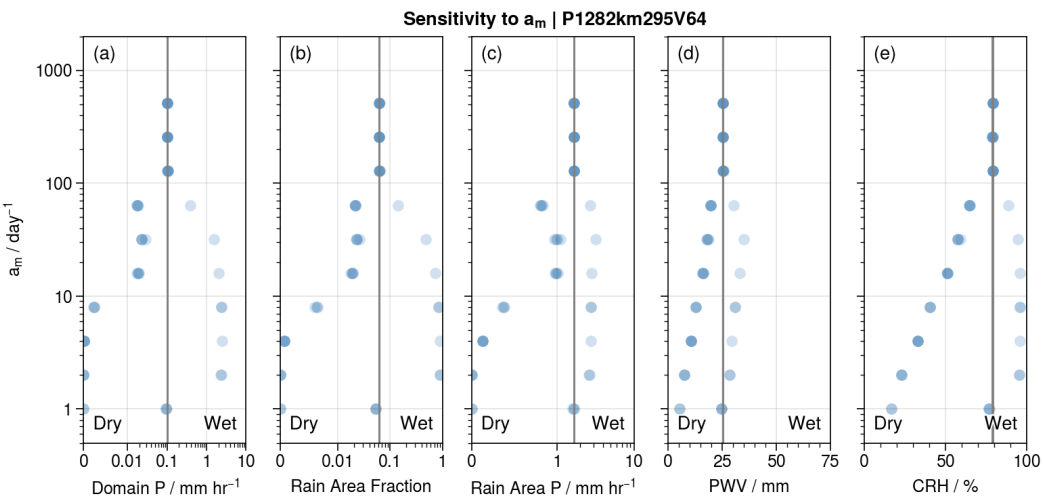

Figure 5. Same scatter plots as in Fig. 4, but this time for an experimental configuration where the sea surface temperature is 295 K (P1282km295V64).

4.3 Sensitivity of Bifurcation to perturbations in Large-Scale Reference profiles

We also test the sensitivity of this bifurcation in model states to perturbations in the large-scale reference profiles of temperature and pressure, and how close they are to RCE. We find that if the large-scale reference profile deviates enough from RCE, then it is possible that our model will be preferentially biased towards one of the two regimes, or that for some a_m, one of the two states may not be observed ($a_m = 8, 32, 128$ for the control configuration). In contrast, for the P1282km295V64 configuration, the large-scale profile was close enough to RCE such that both wet and dry states were obtained for all a_m (Fig. 5) such that the shape is fully outlined.

However, despite the biases that such deviations can cause in the frequency at which one regimes shows up in our results, it is important to note that there are no noticeable differences in the strength of the bifurcation, or on the climatological characteristics of the two regimes at different values of a_m. It is only the frequency at which each of the two regimes show up that changes as a result of these small deviations in the large-scale reference profile. To confirm that this is the case, we perturb the large-scale reference temperature profile of the control configuration (P1282km300V64) by ± 0.05 K uniformly

across all heights. We expect that when we perturb the large-scale reference temperature to be warmer, that the resulting states will be biased towards the dry-regime, and vice versa for a cooler perturbation. We see that this is indeed the case in Fig. 6.

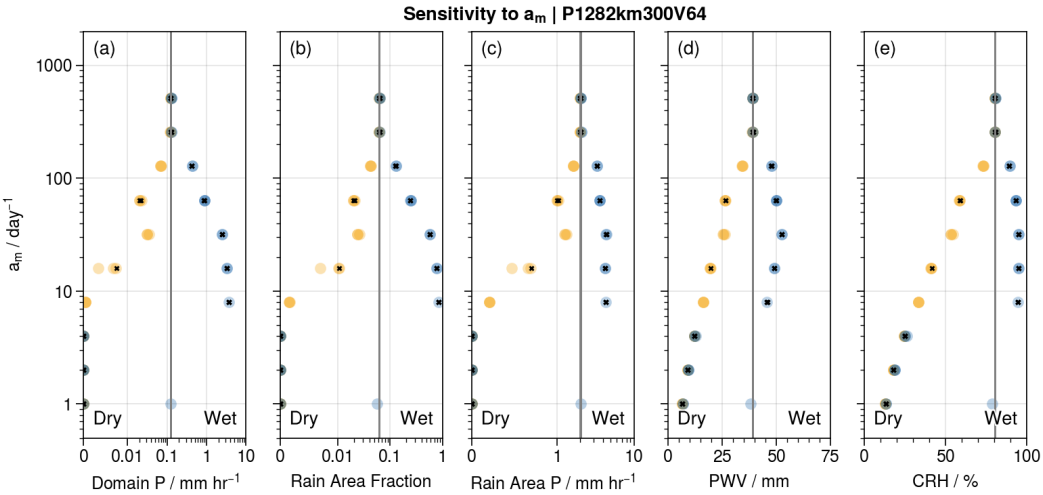

Figure 6. As in Fig. 4, but for model runs with warm (yellow) and cold (blue) perturbations to the large-scale reference profile. The bifurcation with the original, unperturbed reference profile is given in the black crosses. We see that the climatologies of the wet and dry regimes are the same in both perturbed and unperturbed runs, but the frequencies at which each of the regimes appears is different even with small perturbations.

That even such a small perturbation in the temperature profile can cause our final results to collapse into either fully wet, or fully dry, states, implies a huge degree of instability in the RCE state similar to that found by Emanuel et al. (2014). More importantly, this instability must be considered when using weak temperature gradient schemes.

4.4 Interactive Radiation is not a necessary condition for Convective Self-Aggregation

Several studies have shown that the self-aggregation of convection can occur in simulations with static radiation (e.g. Tompkins, 2001; Sessions et al., 2016). However, convective self-aggregation in Tompkins (2001) was not as strong as when interactive radiation was used, and occurred only when surface-flux feedbacks were sufficiently strong. In fact, Daleu et al. (2015) was unable to obtain convective self-aggregation using the DGW implementation that we used. They also found that the time-averaged mean climatology of WTG runs when static radiation was used were not much different from the RCE state. Most of these observations are supported by our model results when static radiation of Pauluis and Garner (2006) was used (Fig. 7), with the bifurcation of our final, time-averaged, model states being much smaller in magnitude compared to model runs with interactive radiation.

However, contrary to several previous studies (Tompkins, 2001; Sessions et al., 2015, 2016), we were able to obtain these bifurcations in our model results without needing to set a threshold for surface-flux feedbacks. Nor is it necessary for us to start from a moist environment to obtain a dryer (if non-precipitating) equilibrium state. Furthermore, many of the abovementioned studies analyzed the time-averaged model output. A closer look at the time-series of column relative humidity (Fig. 8), however, shows us

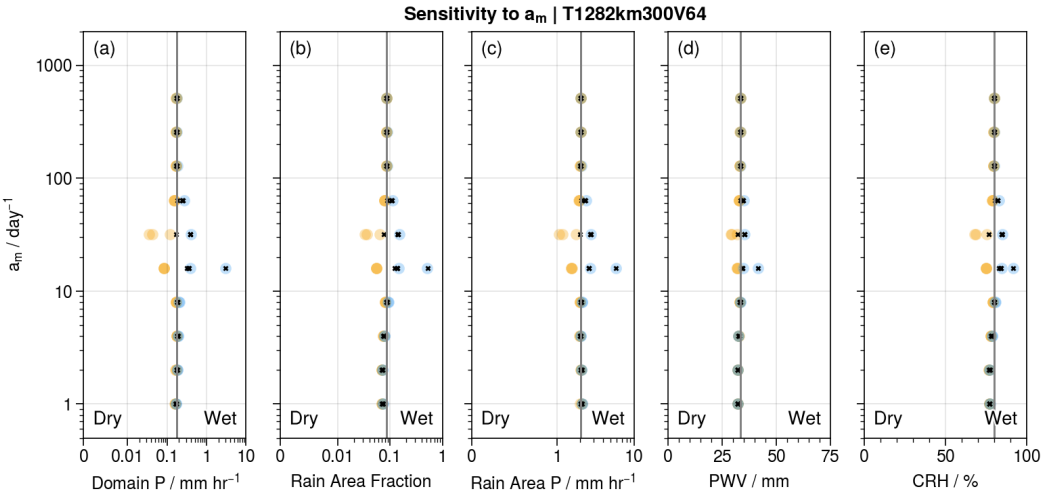

Figure 7. As in Fig. 4, but using a large-scale reference profile that is farther from RCE than our final model run. We see that although the bifurcation in model states is still present and takes on a very similar curve, for some a_m values we are unable to obtain both, but only one of the regimes. In this case, the wet regime is heavily favoured, and the dry regime is not present for $a_m = 32, 128$, though their positions can be extrapolated from other data points.

that in actuality as $a_m \to 0$, the model state does not return to RCE, but instead *fluctuates* between both the dry and the wet regimes (Fig. 8) such that the time-averaged mean climatology is close to RCE. This is very different from models where $a_m \gg 1$, where daily-averaged column relative humidity and precipitation rate show very little variability, and thus indicates a distinct climatology from RCE.

This would indicate that self-aggregation can possibly occur spontaneously in large-domain model runs with static radiation without needing vertical wind-shear. However, the oscillation between the wet and dry regimes would indicate that these areas of self-aggregation are temporally unstable and last on the order of a day as opposed to the semi-stable regions found in large-domain runs with interactive radiation. Lastly, as noted above, (Daleu et al., 2015) concluded that the DGW implementation under static radiation did not produce any noticeable change in model climatology from RCE. However, we have shown that it is possible to find a time-averaged model climatology that appears close to RCE, but is actually not equivalent to RCE. There is no evidence that Daleu et al. (2015) had a look at the time-series of precipitation before drawing their conclusions, and thus we must caution readers that sometimes certain statistical methods can be misleading and lead to incorrect conclusions.

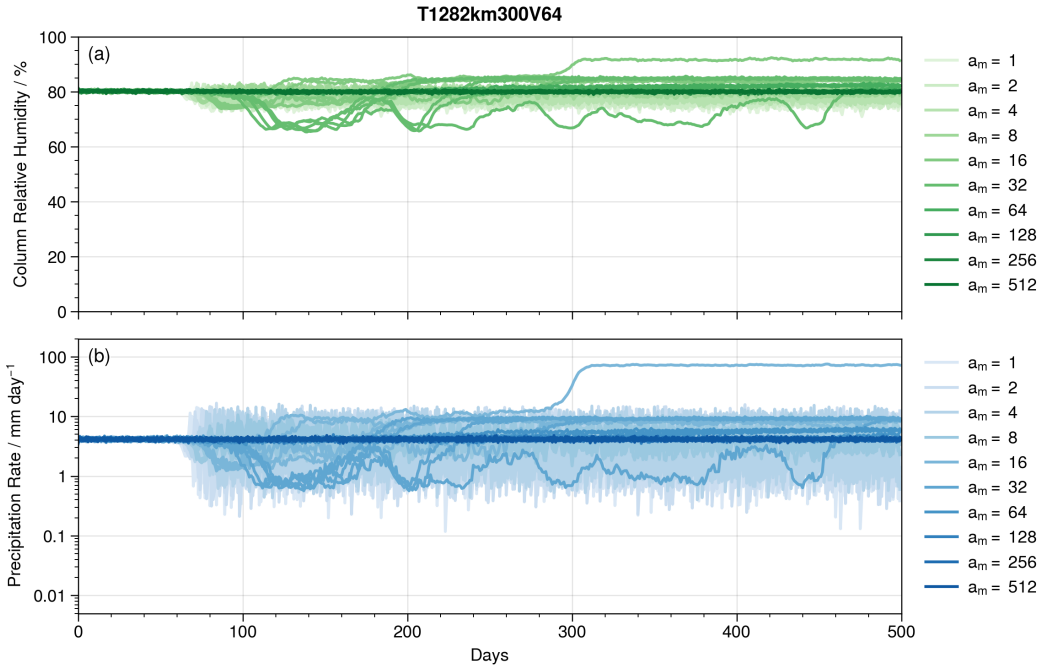

Figure 8. We plot the time-series of (a) column relative humidity and (b) precipitation for DGW runs without interactive radiation (model configuration T1282km300V64). We see that while the time-average of these fields would indicate that the simplified, non-interactive radiative scheme of Pauluis and Garner (2006) suppresses the bifurcation of the model states into dry and wet regimes, the time-series plots indicate that it would be more accurate to state instead that the model is *fluctuating between* these two regimes.

5 Comparison of our results to large-domain simulations

Many studies have considered the abovementioned wet and dry regimes to be analogues of the wet and dry states seen in large-domain RCE simulations when convection undergoes large-scale self-aggregation. However, by varying the strength of the WTG approximation that is applied to our model, in Fig. 4e, we see that our results have the potential to span throughout the entire swathe of column relative humidities (CRH) found in large-domain simulations. We compare our modelling results from the WTG framework against a large-domain simulation run we conducted with the following configuration:

- Sea-surface temperature: 300 K
- Domain size: (a) 4096 × 128 km at 2 km res, with 64 vertical levels
- Radiative Schemes: RRTM Perpetual

Our simulation ran for 100 days, and convection had self-aggregated by the end. However, we note that our simulation took longer to reach a self-aggregated state compared to several other studies, which we believe is a cosmetic difference that can likely be attributed in part to the size of our domain (which is smaller and thus self-aggregation is harder), and the finer horizontal resolution we ran our simulation in. Using the distributions of (a) precipitable water and (b) column relative humidity (Fig. 9) as a gauge, we see that a degree of self-aggregation was visible starting from around day 30, but that at only around day 70 did the dry area(s) grow to a sizeable area. Therefore our model

statistics are taken from day 75 onwards in order to best captured the climatology of self-aggregated convection in our simulation.

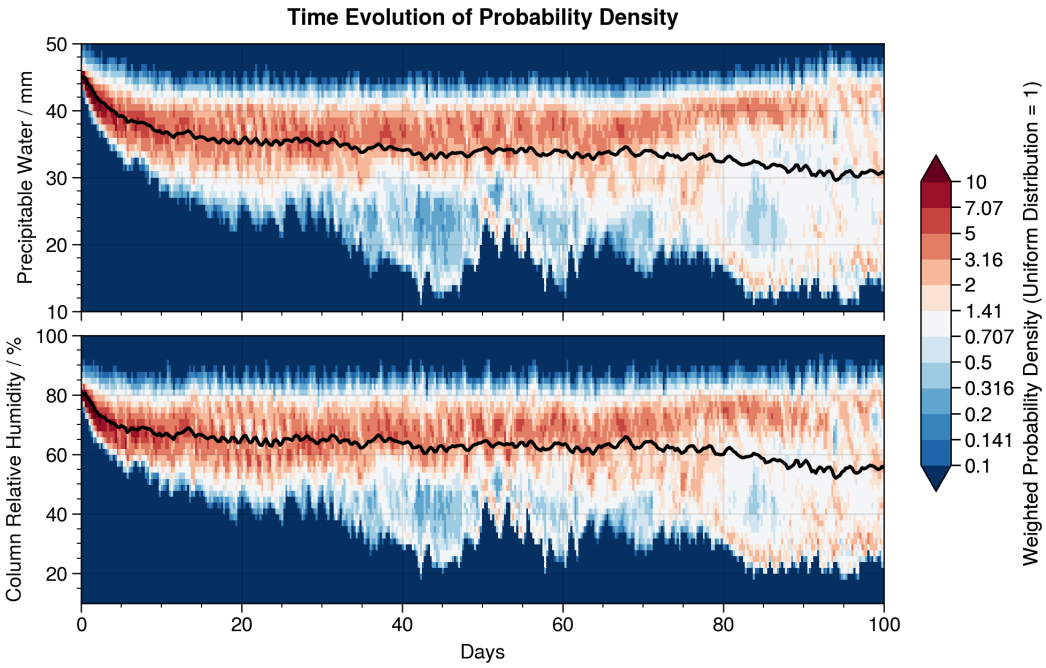

Figure 9. Time-series plots of the probability density (where a uniform distribution is normalized to 1) of (a) preciptiable water and (b) column relative humidity over the simulation. We see that a degree of self-aggregation in convection is attained by day 40, but that dry regimes only become prominent from day 75 onwards.

5.1 Climatology of the Self-Aggregated State

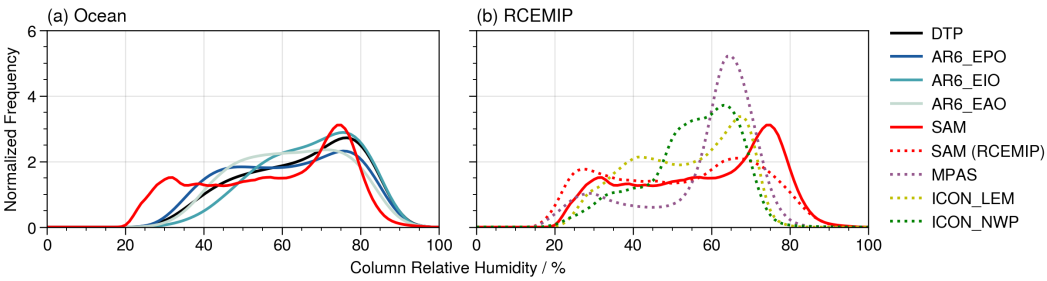

Figure 10. The density distribution of column relative humidity frequencies, normalized such that 1 corresponds to a uniform distribution. We plot the density distribution of our model run (red solid line) against values calculated from (a) ERA5 reanalysis over different AR6 tropical ocean basins and (b) RCEMIP large-domain models at 300 K sea surface temperature.

We start by comparing the distribution of column relative humidity to ERA5 reanalysis output over the tropical ocean regions and RCEMIP large-domain simulations at 300 K. We see that our model run more closely follows the distribution of column rel-

ative humidity in reanalysis output as opposed to RCEMIP simulations, which are overall drier than reanalysis. Column relative humidity over the tropical oceans in ERA5 shows a bimodal distribution, as does our simulation and those in RCEMIP. However, the moist peak in our model run coincides quite closely with that observed over tropical oceans in reanalysis, while in RCEMIP simulations this moist peak is drier overall by about 10%. The dry peak of our simulation however, is also drier by about 10% than in ERA5 while similarly being close to the RCEMIP dry peaks.

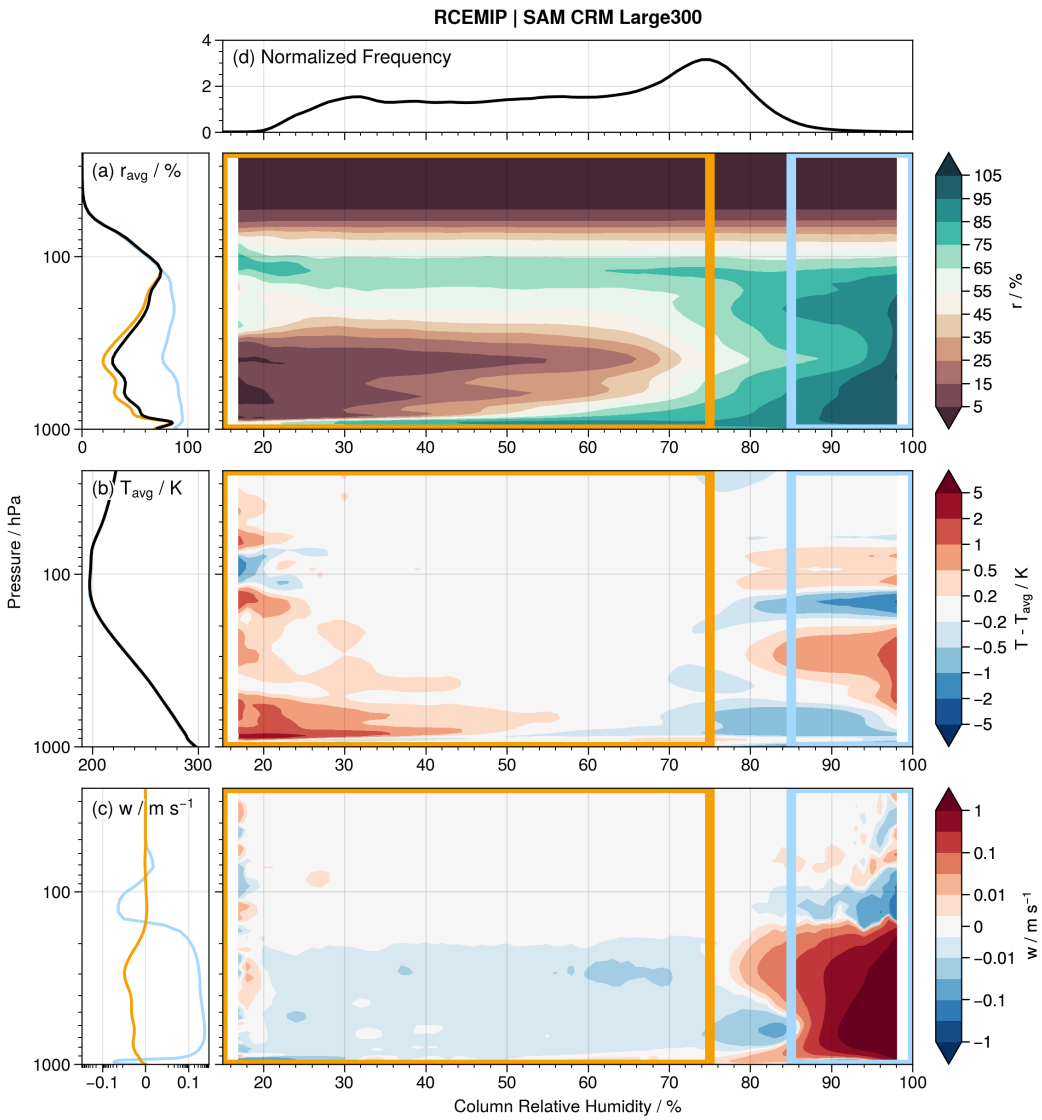

Figure 11. Vertical profiles of (a) relative humidity r, (b) atmospheric temperature T and (c) vertical velocity w for each bin of column relative humidity averaged over the last 25 days of the model simulation. We also take the average for regions of vertical ascent (blue) and subsidence (yellow) in the troposphere that correspond to areas of strong convection and dry downwelling respectively, and plot the vertical profiles in panels on the left. We also plot the (d) normalized distribution of column relative humidity during this time.

Next, we bin the column relative humidity values into bins from 0-100% in steps of 1% and calculate the mean vertical profile of (a) relative humidity, (b) atmospheric temperature and (c) vertical velocity for each of these bins. We plot our results in Fig. 11. We see that the regions of vertical ascent in the troposphere, that correspond to regions of strong convective activity, actually make up a very small portion of the domain - in fact, from Fig. 11d we see that both the peaks of the bimodal distribution occur in regions of subsidence, as opposed to one occurring in regions of ascent and the other in regions of subsidence. This is reflected in the magnitude of the vertical velocity in regions of ascent as opposed to regions of subsidence. Since the domain is periodic, there is no loss of mass over the duration of the model run, and therefore in theory vertical ascent must balance vertical descent such that the domain-mean vertical velocity is zero. Indeed, consistent with our expectations, we see that the magnitude of the vertical velocity in regions of ascent is roughly 100x greater than the magnitudes in the regions of vertical descent.

We also see that regions of ascent generally have a warmer mid-troposphere than regions of subsidence. However, in the lower troposphere, regions of ascent are generally cooler than the domain mean, while regions of subsidence have warmer lower-tropospheres. However, these horizontal perturbations in temperature are very small compared to the variability in temperature with height. In the DGW framework (Eqn. (6)), this implies that the momentum damping is very small, as the response in vertical velocity inversely proportional to a_m.

5.2 Comparison of the vertical profiles of WTG runs to large-domain self-aggregated simulations

We next bin the vertical profiles of the WTG runs by column relative humidity (Fig. 12) and compare it to the results shown in Fig. 11, with binned profiles for the large-domain simulation are plotted as contour lines in Fig. 12. An important point to note is that temperature perturbations for the WTG runs in Fig. 12 are compared against the RCE temperature vertical profile, while temperature perturbations in the large-domain simulation (Fig. 11) are compared against the domain-mean temperature profile. We see from Figs. 4,5 that as $a_m \to 0$, the column relative humidity of the dry regimes decreases monotonically, while the column relative humidity of the moist regimes increases before plateauing near saturation at $\sim 95\%$.

We see that the domain-mean profile of relative humidity for the large-domain simulation is overall significantly lower than RCE, which is to be expected due to the dominance of dry regions in convective self-aggregation and has been corroborated in previous studies studying self-aggregation. However, the average relative humidity profile for atmospheric columns between 75-85% column relative humidity shows a very similar profile to the RCE vertical profile in the lower-mid troposphere, while being significantly drier in the upper troposphere. This is likely due to horizontal advection of moisture away from regions of strong convective activity in the upper troposphere. Nonetheless, the overall evolution of the relative humidity profile from low to high column relative humidity is roughly similar between WTG runs and our large-domain simulation.

We also note that the variability in the vertical profile of temperature with column relative humidity is much more significant in our WTG runs as compared to our large-domain simulations. Perturbations to the mid-tropospheric temperature can reach up to 10 K or even higher in our WTG runs in the moist regime when column relative humidity is high, but for the large-domain simulations the temperature perturbation rarely exceed 1 K, near an order of magnitude difference. However, conversely the WTG-induced vertical velocity in our WTG runs is roughly an order of magnitude smaller than the vertical ascent in regions of strong convective activity in our large-domain simulations. This

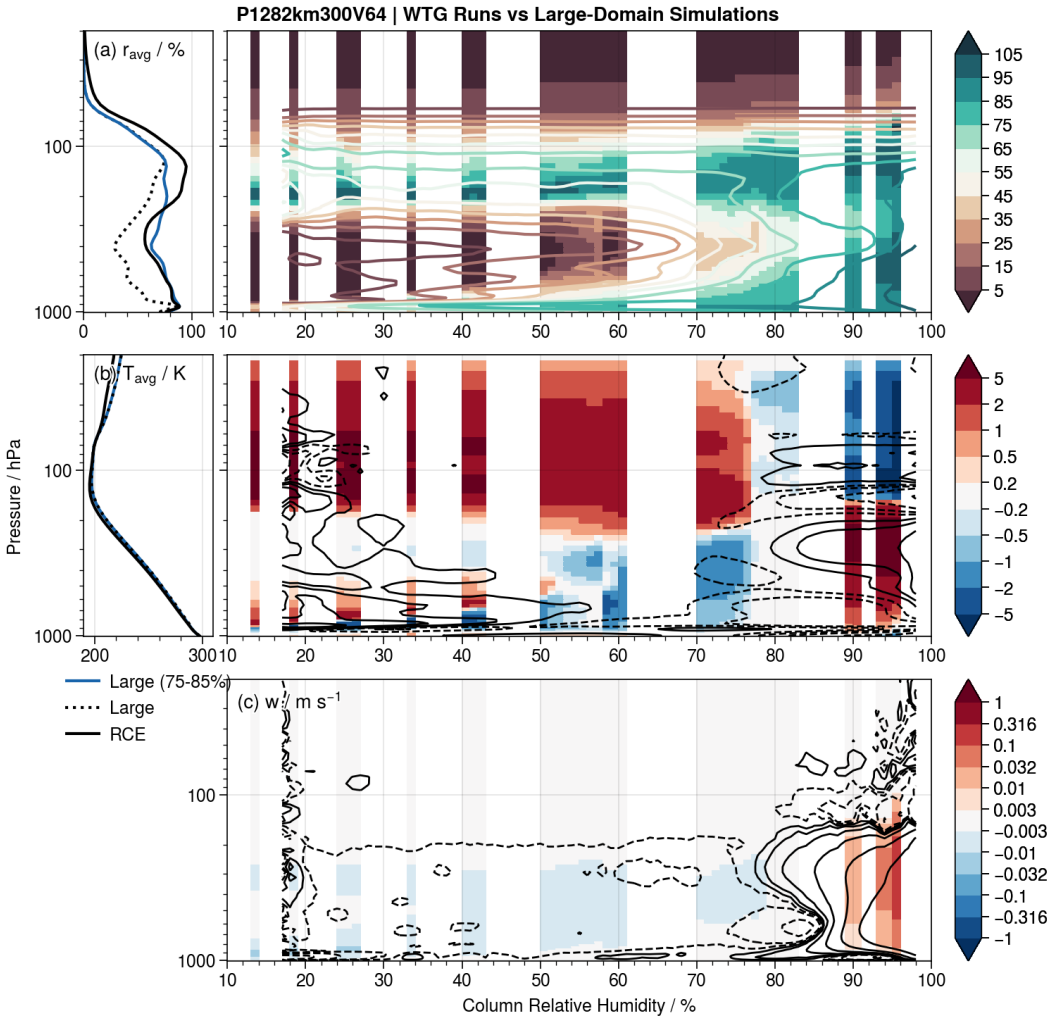

Figure 12. As with Fig. 11, but for the WTG runs, binned by column relative humidity. Results for the large-domain run are shown as contour lines using the same contour levels as the filled meshgrid. Dashed lines indicate negative values. On the left panels we compare the domain-mean vertical profile for the large-domain simulation with the RCE profile used to generate the WTG runs. For (b), the temperature perturbation is against the RCE profile for the WTG runs, and against the large-domain profile for the large-domain run.

would imply that the momentum damping of gravity waves is much weaker (i.e. the weak temperature gradient approximation is stronger) in the large-domain simulation.

Overall, we see that the climatology of the WTG runs does, to some extent, mirror that of self-aggregated large-domain simulations. However, the temperature perturbations required to create such vertical ascent are much stronger in WTG simulations, while the magnitude of the induced vertical winds are much weaker. This could be due to the presence of horizontal advection in large-domain simulations that would further smooth out temperature gradients across the domain, and the convergence of these winds in regions of strong convection could in turn help increase the strength of vertical ascent in these regions.

6 Conclusions

The weak temperature gradient is an important framework that can help us understand tropical dynamics and convective self-aggregation in small-domain models, limiting the necessity of running computationally expensive large-domain simulations over long periods of time. However, we have shown that in configurations with fixed sea surface temperature, small-domain models spun-up to RCE can be very sensitive to the reference temperature and humidity profiles in the large-scale domain when the WTG approximation is applied. Small temperature perturbations in the large-scale profile from RCE can cause the final model state to collapse exclusively into either wet or dry regimes, and therefore small-domain model experiments with fixed sea surface temperature must take this into account.

We also show that it is likely possible to obtain convective self-aggregation in simulations with static radiation. Previous studies (e.g. Daleu et al., 2015) have shown that the time-mean climatology of simulations of static radiation in small domains is close to that of RCE. However, our results have shown that while the time-mean might be similar, the daily variability in the climatology is much greater such that the instantaneous climatology of our simulations can be very different from RCE, which varies very little on a day-to-day basis. To us, this indicates that it is possible for self-aggregation to occur on very short timescales in large-domain simulations with static radiation rather than in semi-stable states when the radiation is interactive.

Lastly, from comparisons of the vertical profiles between our WTG runs and large-domain simulations, we find that the vertical profiles of relative humidity are similar, barring the effects of horizontal advection between regions of convective activity and subsidence. However, we find that the temperature perturbations generated by WTG runs are an order of magnitude greater than in large-domain simulations, while the magnitudes of the vertical winds are lower by an order of magnitude.

Acknowledgments

Enter acknowledgments, including your data availability statement, here.

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Appendix A Factors affecting the time required for models to reach Radiative-Convective Equilibrium

In Section 3.2, we noted that the time taken for the RCE state to reach an equilibrium temperature profile can vary depending on the domain-size, the horizontal and vertical resolutions, and the sea-surface temperature.